

HDOC DAY – 27

Aim

To write a C program that displays a menu to the user and uses a switch statement to print different patterns (Full Pyramid, Inverted Pyramid, Diamond Pattern, Palindromic Number Pyramid, and Repeated-Number Pyramid) based on the user's choice. Each pattern is implemented using nested loops inside a separate function.

Algorithm

- 1. Start the program.**
- 2. Define separate functions** for each pattern option:
 - a. fullPyramid()
 - b. invertedPyramid()
 - c. diamondPattern()
 - d. palindromicPyramid()
 - e. repeatedNumberPyramid()
- 3. In the main() function:**
 - a. Display a menu with 5 pattern choices.
 - b. Read the user's menu choice (choice).
 - c. Use a switch(choice) statement to call the corresponding function based on the input.
 - d. If the user enters an invalid option, print an error message using default.
- 4. Inside each pattern function:**
 - a. Ask the user to enter the number of rows (n).
 - b. Use nested for loops (outer loop for rows, inner loops for printing spaces and numbers/asterisks) to construct the pattern.
- 5. Stop the program.**

C Code

```
#include <stdio.h>
```

```
// Function for Full Pyramid
```

```
void fullPyramid() {
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++) {
```

```
        // print spaces
```

```
        for (j = 1; j <= n - i; j++) {
```

```
            printf(" ");
```

```
        }
```

```
        // print stars
```

```
        for (j = 1; j <= 2 * i - 1; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

```
// Function for Inverted Pyramid
```

```
void invertedPyramid() {
```

```
    int n, i, j;
```

```
printf("Enter number of rows: ");
scanf("%d", &n);

for (i = n; i >= 1; i--) {
    // print spaces
    for (j = 1; j <= n - i; j++) {
        printf(" ");
    }
    // print stars
    for (j = 1; j <= 2 * i - 1; j++) {
        printf("*");
    }
    printf("\n");
}
}
```

```
// Function for Diamond Pattern
void diamondPattern() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    // upper pyramid
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
    }
}
```

```

    }
    for (j = 1; j <= 2 * i - 1; j++) {
        printf("*");
    }
    printf("\n");
}

```

```

// lower pyramid
for (i = n - 1; i >= 1; i--) {
    for (j = 1; j <= n - i; j++) {
        printf(" ");
    }
    for (j = 1; j <= 2 * i - 1; j++) {
        printf("*");
    }
    printf("\n");
}
}

```

// Function for Palindromic Number Pyramid

```

void palindromicPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {

```

```
// print spaces
for (j = 1; j <= n - i; j++) {
    printf(" ");
}

// print numbers 1 to i
for (j = 1; j <= i; j++) {
    printf("%d", j);
}

// print numbers i-1 down to 1
for (j = i - 1; j >= 1; j--) {
    printf("%d", j);
}

printf("\n");
}
}
```

// Function for Repeated-Number Pyramid

```
void repeatedNumberPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        // print spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
    }
}
```

```
    }  
    // print current row number  
    for (j = 1; j <= 2 * i - 1; j++) {  
        printf("%d", i);  
    }  
    printf("\n");  
}  
}
```

```
int main() {  
    int choice;  
  
    printf("1. Full Pyramid\n");  
    printf("2. Inverted Pyramid\n");  
    printf("3. Diamond Pattern\n");  
    printf("4. Palindromic Number Pyramid\n");  
    printf("5. Repeated-Number Pyramid\n");  
    printf("Enter your choice: ");  
    scanf("%d", &choice);  
  
    switch (choice) {  
        case 1:  
            fullPyramid();  
            break;  
        case 2:  
            invertedPyramid();
```

```
        break;
    case 3:
        diamondPattern();
        break;
    case 4:
        palindromicPyramid();
        break;
    case 5:
        repeatedNumberPyramid();
        break;
    default:
        printf("Wrong choice!");
}

return 0;
}
```

CODE SCREENSHOT :

```
#include <stdio.h>
// Function for Full Pyramid
void fullPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        // print spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // print stars
        for (j = 1; j <= 2 * i - 1; j++) {
            printf("*");
        }
        printf("\n");
    }
}

// Function for Inverted Pyramid
void invertedPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--) {
        // print spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // print stars
        for (j = 1; j <= 2 * i - 1; j++) {
            printf("*");
        }
        printf("\n");
    }
}

// Function for Diamond Pattern
void diamondPattern() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    // upper pyramid
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++) {
            printf("*");
        }
        printf("\n");
    }
}
```

```

    }

    // lower pyramid
    for (i = n - 1; i >= 1; i--) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++) {
            printf("*");
        }
        printf("\n");
    }
}

// Function for Palindromic Number Pyramid
void palindromicPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        // print spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // print numbers 1 to i
        for (j = 1; j <= i; j++) {
            printf("%d", j);
        }
    }
}

```

```
int main() {
    int choice;

    printf("1. Full Pyramid\n");
    printf("2. Inverted Pyramid\n");
    printf("3. Diamond Pattern\n");
    printf("4. Palindromic Number Pyramid\n");
    printf("5. Repeated-Number Pyramid\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            fullPyramid();
            break;
        case 2:
            invertedPyramid();
            break;
        case 3:
            diamondPattern();
            break;
        case 4:
            palindromicPyramid();
            break;
        case 5:
            repeatedNumberPyramid();
            break;
        default:
```

OUTPUT SCREENSHOT :

manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./27q

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid

Enter your choice: 1

Enter number of rows: 5

```
  *
 ***
*****
*****
*****
*****
```

[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./27q

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid

Enter your choice: 2

Enter number of rows: 5

```
*****
*****
****
***
**
*
```

[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./27q

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid

Enter your choice: 3

Enter number of rows: 5

```
  *
 ***
*****
*****
*****
*****
*****
****
***
**
*
```

manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % █

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./27q
```

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid

Enter your choice: 4

Enter number of rows: 5

```
  1
 121
12321
1234321
123454321
```

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./27q
```

1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid

Enter your choice: 5

Enter number of rows: 5

```
  1
 222
33333
4444444
555555555
```

```
manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % █
```